

Aufgabe 1

a) (i) $(x_n) = (0, 3, 6, 9, 12, \dots) \Rightarrow f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto 3n$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ 3 \cdot 0 & 3 \cdot 1 & 3 \cdot 2 & 3 \cdot 3 & 3 \cdot 4 \end{matrix}$

(ii) $(x_n) = (-4, -1, 2, 5, 8, \dots) \Rightarrow f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto -4 + 3n$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \\ -4 & -4+3 & -4+3 \cdot 2 & -4+3 \cdot 3 \end{matrix}$

(iii) $(x_n) = (0, -1, 2, -3, 4, \dots) \Rightarrow f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto (-1)^n \cdot n$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ 1 \cdot (-1)^0 & 2 \cdot (-1)^1 & 3 \cdot (-1)^2 \end{matrix}$

(iv) $(x_n) = (0, 1, -2, 3, -4, \dots) \Rightarrow f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto (-1)^{n+1} \cdot n$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ 1 \cdot (-1)^1 & 2 \cdot (-1)^2 & 3 \cdot (-1)^3 \end{matrix}$

(v) $(x_n) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots\right)$

$(x_n) = \left(\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \frac{1}{2^5}, \dots\right) \Rightarrow f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto \frac{1}{2^{n+1}}$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \\ 0 & 1 & 2 & 3 \end{matrix}$

(vi) $(x_n) = \left(1, \frac{1}{3}, \frac{1}{7}, \frac{1}{15}, \frac{1}{31}, \dots\right)$

$(x_n) = \left(\frac{1}{2^1-1}, \frac{1}{2^2-1}, \frac{1}{2^3-1}, \frac{1}{2^4-1}, \frac{1}{2^5-1}, \dots\right)$

$\Rightarrow f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto \frac{1}{2^{n+1}-1}$

b) $a_0 = 2, a_{n+1} = \frac{2a_n}{2+a_n}$

$a_1 = \frac{2a_0}{2+a_0} = \frac{2 \cdot 2}{2+2} = 1$

$a_2 = \frac{2a_1}{2+a_1} = \frac{2 \cdot 1}{2+1} = \frac{2}{3}$

$a_3 = \frac{2a_2}{2+a_2} = \frac{2 \cdot \frac{2}{3}}{2+\frac{2}{3}} = \frac{4}{8} = \frac{1}{2}$

Aufgabe 2

a) mindestens x_n divergiert // y_n divergiert

$(x_n): \text{divergiert} \quad \left\{ \begin{array}{l} (x_n) + (y_n): \text{konvergiert?} \\ (y_n): \text{divergiert} \end{array} \right.$

$|x_n| = n \Rightarrow (x_n) + (y_n) = 0 \Rightarrow \text{konvergiert auf } 0$

(y_n) ...

$$(x_n) = n \Rightarrow (x_n) + (y_n) = 0 \Rightarrow \text{Konvergiert auf } 0$$

$$(y_n) = -n$$

b) $(x_n) \cdot (y_n)$: Konvergiert

$$\begin{aligned} (x_n) &= n \quad (\text{divergiert}) \\ (y_n) &= \frac{1}{n} \quad (\text{Konvergiert}) \end{aligned} \Rightarrow (x_n) \cdot (y_n) = 1$$

$$\Rightarrow \text{Konvergiert auf } 1$$

c) $(x_n) = \{0, 0, 0, \dots\} \Rightarrow x_n < y_n \text{ or } 0 < \frac{1}{n} \text{ mit } n \in \mathbb{N}$

$$(y_n) = \left(\frac{1}{n}\right)^2$$

$$\Rightarrow \lim_{n \rightarrow \infty} (x_n) = 0 = \lim_{n \rightarrow \infty} (y_n) = \lim_{n \rightarrow \infty} \frac{1}{n^2}$$

$$\uparrow \frac{1}{n} : n \rightarrow \infty, \frac{1}{n} \rightarrow 0$$

d) $(x_n) = (-1)^n$ x_n : divergiert

$$|x_n| = |(-1)^n| = 1^n \rightarrow \text{Konvergiert}$$

Aufgabe 3

Bestimmen Sie den Grenzwert der Folge $(x_n) \subseteq \mathbb{R}$, falls

(a) $x_n = \frac{3n^2 + 4n + 20}{4n^3 + 1000}$, (b) $x_n = \frac{2n^3 + 7n^2 + 12}{-5n^3 - n + 3}$, (c) $x_n = \left(2 + \frac{3}{n}\right)^5$,
(d) $x_n = \sqrt[n]{21000}$, (e) $x_n = \left(\frac{n+1}{n}\right)^n$, (f) $x_n = (-1)^n \frac{\sin(n) \cos(n)}{3n^2}$

a) $x_n = \frac{3n^2 + 4n + 20}{4n^3 + 1000} = \frac{n^3 \left(\frac{3}{n} + \frac{4}{n^2} + \frac{20}{n^3} \right)}{n^3 \left(4 + \frac{1000}{n^3} \right)} = \frac{\frac{3}{n} + \frac{4}{n^2} + \frac{20}{n^3}}{4 + \frac{1000}{n^3}}$

$$\lim_{n \rightarrow \infty} \left(\frac{\frac{3}{n} + \frac{4}{n^2} + \frac{20}{n^3}}{4 + \frac{1000}{n^3}} \right) = \frac{0 + 0 + 0}{4 + 0} = 0$$

hochsten Exponenten

$\uparrow 0, n \rightarrow \infty$ $\uparrow 0, n \rightarrow \infty$ $\uparrow 0, n \rightarrow \infty$

$\uparrow 0, n \rightarrow \infty$ $\uparrow 0, n \rightarrow \infty$

b) $x_n = \frac{2n^3 + 7n^2 + 12}{-5n^3 - n + 3} = \frac{n^3 \left(2 + \frac{7}{n} + \frac{12}{n^2} \right)}{n^3 \left(-5 - \frac{1}{n^2} + \frac{3}{n^3} \right)} = \frac{2 + \frac{7}{n} + \frac{12}{n^2}}{-5 - \frac{1}{n^2} + \frac{3}{n^3}}$

$$\lim_{n \rightarrow \infty} \left(\frac{2 + \frac{7}{n} + \frac{12}{n^2}}{-5 - \frac{1}{n^2} + \frac{3}{n^3}} \right) = \frac{2 + 0 + 0}{-5 - 0 + 0} = \frac{2}{-5} = -\frac{2}{5}$$

$\uparrow 0, n \rightarrow \infty$ $\uparrow 0, n \rightarrow \infty$ $\uparrow 0, n \rightarrow \infty$

$\uparrow 0, n \rightarrow \infty$ $\uparrow 0, n \rightarrow \infty$

c) $\lim_{n \rightarrow \infty} (x_n) = \lim_{n \rightarrow \infty} \left(2 + \frac{3}{n} \right)^5 = (2 + 0)^5 = 2^5$

$\uparrow 0, n \rightarrow \infty$

$$d) \lim_{n \rightarrow \infty} 2^n \sqrt[n]{2^{1000}} = \lim_{n \rightarrow \infty} (2^{1000})^{1/2n} = \lim_{n \rightarrow \infty} 2^{\frac{1000}{2n}} = \lim_{n \rightarrow \infty} 2^{\frac{500}{n}} = \lim_{n \rightarrow \infty} \sqrt[n]{2^{500}} = 1$$

Rule $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$

$$e) x_n = \left(\frac{1+n}{n} \right)^n = \left(\frac{1}{n} + 1 \right)^n$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n} + 1 \right)^n = e$$

$$f) x_n = (-1)^n \cdot \frac{\sin(n) \cdot \cos(n)}{3n^3} = (-1)^n \cdot \frac{\sin(2n)}{2 \cdot 3n^3}$$

Wegen $-1 \leq \sin(n) \leq 1 \Rightarrow 0 \leq |\sin(2n)| \leq 1$

$$0 \leq |x_n| = \left| \frac{(-1)^n \cdot \sin(2n)}{2 \cdot 3n^3} \right| \leq \frac{1}{2 \cdot 3n^3}$$

$$\lim_{n \rightarrow \infty} \frac{1}{2 \cdot 3n^3} = 0 \Rightarrow \lim_{n \rightarrow \infty} x_n = 0$$

Aufgabe 4

$$g) \left| \frac{1}{\sqrt{n}} - 0 \right| < \varepsilon \Rightarrow \left(\left| \frac{1}{\sqrt{n}} - 0 \right| \right)^2 < \varepsilon^2$$

$$\Rightarrow \frac{1}{n} < \varepsilon^2 \Rightarrow n > \frac{1}{\varepsilon^2}$$

$$\varepsilon = 10 \Rightarrow n > \frac{1}{100} \quad \text{BS: } n = 1 \quad n \in \mathbb{N}$$

$$\varepsilon = 1 \Rightarrow n > 1 \quad \text{BS: } n = 2$$

$$\varepsilon = 1 \Rightarrow n > 1 \quad B_S: n = 2$$

$$\varepsilon = 1/10 \Rightarrow n > 100 \quad B_S: n = 101$$

$$\varepsilon = 1/10^6 \Rightarrow n > 10^{12} \quad B_S: n = 10^{12} + 1$$

b) $|x_n - 0| < \varepsilon \Rightarrow |x_n| < \varepsilon$

Aus der a) $\Rightarrow$ Wenn $n \rightarrow \infty$, $\varepsilon = 0$

$n \rightarrow 0$, $\varepsilon = \infty$

$$\lim_{n \rightarrow \infty} (x_n) = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) = 0$$

$n \rightarrow \infty$, $\varepsilon \rightarrow 0$

Aufgabe 5

$$a) \sum_{k=0}^5 \left(\frac{1}{2} \right)^k = \frac{1 - \left(\frac{1}{2} \right)^6}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{64}}{\frac{1}{2}} = \frac{63}{32}$$

$$b) \sum_{k=1}^5 \left(\frac{1}{2} \right)^k = \left[\sum_{k=0}^5 \left(\frac{1}{2} \right)^k \right] - \left(\frac{1}{2} \right)^0 = \frac{63}{32} - 1 = \frac{31}{32}$$

$$c) \sum_{k=1}^5 \left(\frac{1}{2} \right)^k = \left[\sum_{k=0}^5 \left(\frac{1}{2} \right)^k \right] - 1 = \frac{63}{32} - 1 = \frac{31}{32}$$

$$c) \sum_{k=2}^5 \left(\frac{1}{2}\right)^k = \left[\sum_{k=1}^5 \left(\frac{1}{2}\right)^k \right] - \left(\frac{1}{2}\right)^1 = \frac{31}{32} - \frac{1}{2} = \frac{15}{32}$$

$$d) \sum_{k=0}^{10} (-1)^k = \frac{1 - (-1)^{11}}{1 - (-1)} = \frac{2}{2} = 1$$

$$e) \sum_{k=0}^{11} (-1)^k = \left[\sum_{k=0}^{10} (-1)^k \right] + (-1)^{11} = 1 - 1 = 0$$

$$f) \sum_{k=0}^2 3^k = \frac{1 - 3^3}{1 - 3} = \frac{-26}{-2} = 13$$